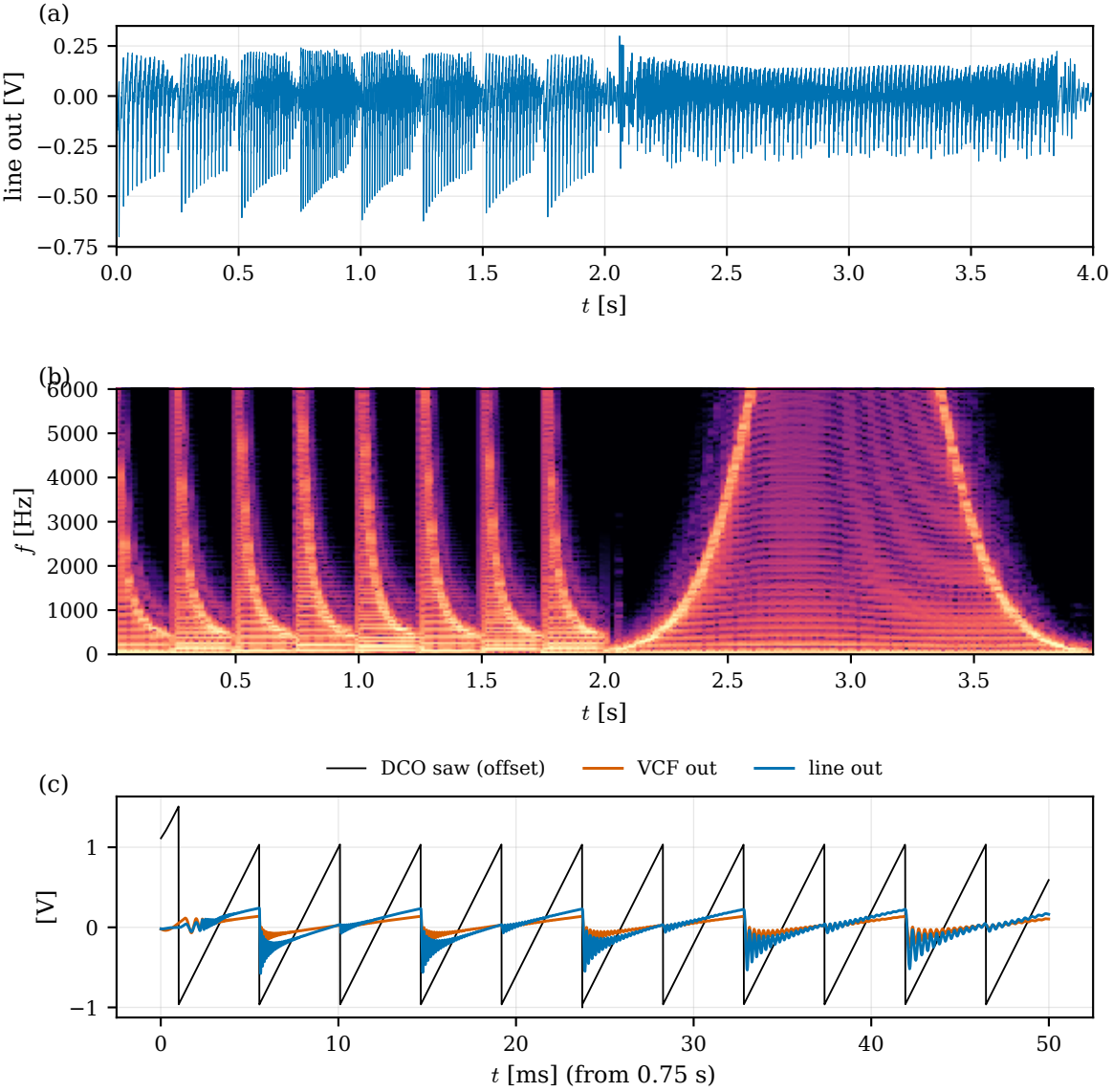

(a)
line out [V]
0.25
0.00
−0.25
−0.50
−0.75
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
t [s]
(b)
f [Hz]
6000
5000
4000
3000
2000
1000
0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
t [s]
DCO saw (offset)
VCF out
line out
(c)
[V]
1
0
−1
0
10
20
30
40
50
t [ms] (from 0.75 s)